

How LEDs Are Being Used In Health Sector

LEDs have proven their versatility in the healthcare sector. Coloured light, for example, has a calming effect on patients, as dynamically changing the light provides the human organism with activating impetus



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LEDs were once almost exclusively used as status indicators on circuit boards or control panels. Today, these can be seen almost everywhere. Small size, low current and low energy consumption of LEDs have led to uses including large-area LED displays for stadium displays and small, thin, lightweight uses such as status indicators and glow lights. It would not be an exaggeration to say that LEDs have already transformed the luminance industry outlook towards everyday lighting devices as well as vehicle lights.

Going beyond the lighting applications of LEDs, their utility has been gaining ground in the health sector, too. This article highlights in brief the developments of

LED-based technologies in terms of medical applications.

Medical applications

Some applications of LEDs in the medical field are discussed here.

Surgical lighting. In comparison to traditional surgical lighting systems using halogen or gas-discharge bulbs, present medical facilities are taking advantage of the inherent characteristics of high-performance LEDs, including superior lighting, longer lives and reduced power consumption. LED-based surgical lights have already become the technology of choice for overhead illumination in operating rooms (ORs) of many hospitals and dental operatories.

High-performance LEDs are putting surgical lighting and light-applied medical treatments on the cutting edge with high luminous outputs and energy-efficient designs. It is due to these reasons that LED surgical lights produce a high intensity white light with high colour temperature for optimum visibility and illumination, which reduce eye strain and mental fatigue of the surgical staff, thus promoting high concentration and energy levels through the duration of medical procedures.

Advanced LED surgical lighting systems typically make use of modular panel designs with each panel equipped with arrays of high-powered LEDs to provide a superior uniform light. These systems use innovative colour-control technology that helps adjust the colour temperature of the output light while maintaining high colour rendering. This allows optimum surface contrast and enhanced tissue differentiation at the surgical site.

Units are thin and lightweight so they



LED-based surgical lights (Credit: www.surgiris.com)